COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	
Programme:	
Course Code & Course Name	
Academic Year / Batch	
Faculty Name	
Assignment Title	
Date of Issue	
Date of Submission	
Maximum Marks	100
Course Outcome(s) – CO	
Bloom's Taxonomy Level	
SDG Mapping (SDG 1-17)	
Industry / Societal Relevance	

B. Assignment Problem / Challenge

Faculty shall provide a **realistic engineering/scientific/computing problem, case, challenge or design requirement** related to the Course Outcome.

The assignment should preferably require students to:

- Apply course concepts rather than reproduce theory.
- Identify and analyze the problem.
- Consider requirements and constraints.
- Develop or compare alternative solutions where appropriate.
- Use relevant modern engineering/computing/design tools.

- Interpret results.
- Make and justify engineering decisions.
- Consider appropriate technical, economic, environmental, societal, ethical, safety or sustainability factors wherever relevant.

C. Problem Statement

Problem / Case / Design Challenge:

[Faculty to provide the detailed assignment problem here.]

D. Requirements and Constraints

Students shall identify or be provided with relevant requirements and constraints.

Examples include:

- Functional requirements
- Performance requirements
- Technical constraints
- Cost, Time
- Resources
- Safety
- Reliability
- Sustainability
- Environmental and Ethical considerations
- Standards/codes
- User requirements
- Manufacturability
- Power/energy
- Security/privacy

Only constraints relevant to the particular course/problem need to be considered.

E. Student Work

1. Problem Understanding and Formulation

Explain:

- What is the problem?
- What are the expected outcomes?
- What information/data is available?
- What assumptions are made?

- What constraints must be satisfied?

2. Application of Course Knowledge

Identify and apply the relevant:

- Principles
- Theories
- Mathematical models
- Algorithms
- Engineering concepts
- Scientific concepts
- Design methodologies

Show necessary calculations, derivations or reasoning.

3. Solution / Design / Methodology

Develop the proposed solution. Depending upon the nature of the course, this may include:

- Design
- Algorithm
- Model
- Circuit
- Architecture
- Experiment
- Process
- Prototype
- Simulation
- Case analysis
- Data analysis
- Mathematical solution

Where appropriate, consider **more than one possible solution**.

4. Use of Modern Tools

Use appropriate tools wherever relevant.

Examples:

CSE/IT: Python, MATLAB, TensorFlow, cloud platforms, Git, simulation tools

ECE/EEE: MATLAB/Simulink, Cadence, Synopsys, Vivado, Multisim, LTspice

Mechanical: ANSYS, SolidWorks, CATIA, MATLAB

Civil: STAAD.Pro, ETABS, AutoCAD, Revit

Biomedical: MATLAB, LabVIEW, signal/image-processing tools

Biotechnology: Bioinformatics/data-analysis/simulation tools

Students shall provide appropriate evidence of tool usage and outputs.

5. Results and Validation

Present the results using appropriate:

- Tables
- Graphs
- Simulation outputs
- Experimental observations
- Screenshots
- Performance metrics
- Statistical analysis
- Test results

Validate whether the proposed solution satisfies the stated requirements.

6. Analysis and Engineering Decision

Interpret the results.

Where applicable:

- Compare alternatives.
- Identify advantages and limitations.
- Discuss trade-offs.
- Explain why the final solution was selected.
- Justify the decision using quantitative or qualitative evidence.

7. Broader Considerations

Where relevant to the problem, discuss the impact of the proposed solution with respect to:

- Sustainability
- Environment
- Society
- Safety
- Ethics
- Economics
- Accessibility

- Professional responsibility

8. Conclusion

Summarize:

- Proposed solution
- Major findings
- Achievement of requirements
- Limitations
- Possible improvements

9. Student Reflection

- What did you learn from this assignment beyond what was directly taught in the classroom?
- If you were given additional time/resources, what would you improve and why?

10. References

Use an appropriate citation format and acknowledge all external sources, datasets, standards, software and AI-assisted tools used.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	**100**

G. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation			L3/L4	10
Application of knowledge			L3/L4	20
Solution / Design / Implementation			L4/L5/L6	20
Modern tool usage			L3/L4	10
Validation			L4/L5	15
Analysis & justification			L4/L5	15
Broader considerations			L4/L5	5
Documentation & reflection			L3/L4	5

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is **not required to map to every PO**.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	≥ 70%
Level 2	60–69%
Level 1	50–59%
Level 0	< 50%

Target CO Attainment: ___________

Actual CO Attainment: ___________

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering:

Documents to be Maintained

The department/course faculty shall maintain:

1. Assignment question/problem statement
2. CO–PO and Bloom's mapping
3. Assessment rubric
4. Student submissions
5. Tool/simulation/experimental evidence, where applicable
6. Evaluated scripts/reports
7. Marks and rubric-wise assessment
8. CO attainment analysis
9. Sample student work representing different performance levels
10. Faculty analysis and continuous-improvement action